

KHON KAEN, Thailand

WMO: 483810

Lat: 16.467N

Lon: 102.784E

Elev: 204

StdP: 98.90

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.7	16.1	7.1	6.4	20.8	9.2	7.4	21.6	6.5	24.2	6.0	25.0	2.0	10	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.5	38.2	24.6	37.1	24.6	36.0	24.7	27.6	32.9	27.1	32.3	26.7	31.7	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	22.2	29.4	26.0	21.9	29.2	25.7	21.5	28.8	89.3	32.9	87.2	32.3	85.5	32.1	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.8	5.1	4.5	DB		11.5	39.8	1.9	1.1	10.1	40.6	9.0	41.2	7.9	41.8	6.5	42.6
			WB		9.7	28.6	1.8	0.7	8.4	29.1	7.4	29.6	6.4	30.0	5.1	30.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	24.2	26.3	28.8	30.2	29.6	29.3	28.5	28.2	27.8	27.3	26.0	24.0
	DBStd	2.84	2.61	2.97	2.84	2.22	1.76	1.37	1.36	1.21	1.14	1.33	2.03	2.63
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	1	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6396	439	457	583	607	609	579	574	563	535	537	479	434
	CDD18.3	3356	181	223	325	357	351	329	316	304	285	278	229	177
	CDH23.3	36656	1633	2391	4011	4629	4209	3918	3478	3151	2782	2737	2163	1554
	CDH26.7	16159	628	1138	2154	2553	2038	1766	1378	1138	946	1015	849	557
Wind	WSAvg	1.9	1.9	1.8	1.8	1.7	1.7	2.0	2.1	2.0	1.6	1.8	2.1	2.2
Precipitation	PrecAvg	1167	7	13	34	63	159	166	161	198	244	98	21	4
	PrecMax	1462	63	77	76	139	381	467	254	301	422	195	102	27
	PrecMin	778	0	0	0	4	18	39	40	98	106	4	0	0
	PrecStd	186	17	21	22	41	103	112	62	56	86	63	33	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	37.2	39.5	40.2	38.4	36.9	36.0	34.8	34.1	34.1	34.9	34.0
		MCWB	22.1	22.6	22.9	24.9	26.2	25.9	25.8	26.3	26.4	25.6	24.3	23.2
	2%	DB	33.0	35.9	38.0	38.8	36.9	35.2	34.1	33.4	33.0	32.9	33.2	32.4
		MCWB	22.2	22.6	23.3	25.0	26.0	25.9	25.8	26.1	26.1	25.1	24.2	23.0
	5%	DB	31.8	34.5	36.7	37.2	35.3	34.1	33.1	32.6	32.1	32.1	32.1	31.1
		MCWB	21.9	22.8	23.6	25.2	25.9	25.9	25.9	25.9	25.7	24.6	23.8	22.3
	10%	DB	30.2	32.9	35.2	35.9	34.0	33.1	32.0	31.6	31.1	31.2	31.0	29.9
		MCWB	21.3	22.8	23.8	25.3	26.0	25.9	25.7	25.8	25.7	24.4	23.2	21.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.0	26.1	27.1	28.1	28.2	27.7	27.5	27.6	27.6	27.0	26.5	25.7
		MCDB	30.7	32.5	34.6	35.5	34.1	32.6	32.0	31.9	31.4	31.1	31.7	30.6
	2%	WB	24.0	25.1	26.2	27.1	27.3	27.1	26.8	26.9	27.0	26.3	25.6	24.2
		MCDB	30.0	32.0	33.8	34.1	33.2	32.2	31.4	31.1	30.7	30.3	30.6	29.7
	5%	WB	23.1	24.4	25.4	26.5	26.8	26.7	26.4	26.5	26.5	25.9	25.0	23.3
		MCDB	29.1	30.9	32.7	33.4	32.5	31.6	30.8	30.5	30.1	29.8	29.7	29.3
	10%	WB	22.3	23.7	24.8	26.0	26.4	26.3	26.1	26.1	26.1	25.5	24.4	22.5
		MCDB	28.2	30.1	31.7	33.0	31.8	31.1	30.3	29.9	29.5	29.1	28.9	28.5
Mean Daily Temperature Range	5% DB	MDBR	12.1	12.1	11.4	10.5	8.8	8.0	7.3	7.1	7.0	8.2	10.1	11.4
		MCDBR	13.1	13.5	13.3	12.3	10.6	9.4	8.9	8.5	8.3	9.1	10.9	12.0
		MCWBR	4.8	4.2	3.9	3.5	3.2	2.8	2.8	2.9	2.9	2.9	3.8	4.4
	5% WB	MCDBR	11.4	11.4	11.5	10.9	9.4	8.3	7.8	7.7	7.5	7.9	9.7	11.3
		MCWBR	4.7	4.2	4.2	3.8	3.3	3.0	3.0	3.1	3.1	3.0	3.9	4.8
Clear-Sky Solar Irradiance	taub	0.475	0.582	0.651	0.612	0.506	0.458	0.446	0.456	0.500	0.536	0.473	0.468	
	taud	2.087	1.822	1.702	1.838	2.161	2.317	2.348	2.307	2.172	2.032	2.181	2.130	
	Ebn at Noon	808	738	703	733	804	835	846	846	811	768	806	802	
	Edh at Noon	156	212	246	215	154	130	126	133	151	170	141	146	
All-Sky Solar Radiation	RadAvg	4.86	5.18	5.29	5.70	5.49	5.15	4.79	4.68	4.60	4.91	4.89	4.72	
	RadStd	0.29	0.26	0.27	0.23	0.29	0.27	0.33	0.28	0.31	0.36	0.18	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	+0.30	N/A	N/A	N/A	+0.17	+0.14	N/A	N/A	+120	+122

Nomenclature: See separate page